

CLIENT DELIVERABLE

SAOS Architecture Overview v3.0

Systack Documentation

ROLE	COLOR	HEX
Headers, CTAs	Navy	 #001a2d
Navy Light	Navy Light	 #002845
Secondary accents	Teal	 #007da9
Primary buttons, links	Cyan	 #00a1db
Gradients	Cyan Bright	 #00c5e0
Backgrounds	Gray 50	 #f8fafc
Cards	Gray 100	 #f1f5f9
Borders	Gray 200	 #e2e8f0
Muted text	Gray 400	 #94a3b8
Body text	Gray 600	 #475569
Headings	Gray 800	 #1e293b
Success	Green	 #22c55e
Error	Red	 #ef4444
Accent highlights	Purple	 #8b5cf6

SAOS Architecture Overview

Document ID: SYS-ARCH-v3.0 **Version:** 3.0 **Status:** SYSTACK INTERNAL **Prepared for:** Systack Engineering **Prepared by:** SOL + ORACLE **Builder:** SOL + ORACLE + ASSEMBLY **Source System:** SAOS v2026.06 **Date:** June 25, 2026 **Support:** Internal

SAOS System Architecture

Overview

SAOS (Systack Agent Orchestration System) is a 10-agent AI fleet deployed on dedicated VPS instances, orchestrated via PostgreSQL task queues and accessible through Tailscale-secured networks.

1. Fleet Composition

10-Agent Canonical Fleet

TIER	AGENT	EMOJI	ROLE	CAPABILITIES
Execution	SOL	👤	Orchestrator	orchestration, execution, synthesis
Execution	CODY	💻	Build Engine	coding, voice, streaming
Execution	ASSEMBLY	🔧	Deployment	n8n, workflows, credentials
Quality/Risk	VALI	✅	Validation	testing, validation, quality
Quality/Risk	PESSI	⚠️	Risk Analysis	security, validation, risk
Intelligence	ORACLE	🧠	Design/Architecture	design, research, planning
Intelligence	ATLAS	📖	Knowledge	memory, documentation
Engagement	CHATTY	💬	Communication	communication, onboarding, content
Engagement	GENI	🎨	Creative	image_gen, video_gen, creative
Compliance	JURIS	⚖️	Legal/Compliance	legal, regulatory oversight

System Loop (Canonical)

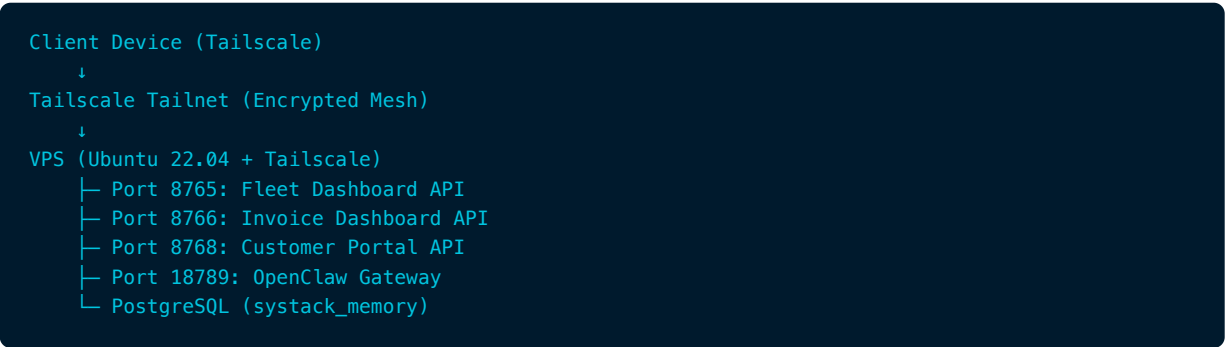
ORACLE → Design → CODY → Build → ASSEMBLY → Deploy → VALI → Validate → PESSI → Stress-test → SOL → Execution

2. Infrastructure

VPS Tiers

PLAN	VULTR TIER	VCPU	RAM	STORAGE	MONTHLY COST
Business	vhp-8c-16gb-amd	8	16GB	160GB NVMe	\$96
Enterprise	voc-g-8c-32gb-160s-amd	8	32GB	160GB NVMe	\$240

Network Architecture



Persistence

SERVICE	PORT	LAUNCHAGENT	STATUS
Fleet Dashboard	8765	net.systack.dashboard	✅ Running

SERVICE	PORT	LAUNCHAGENT	STATUS
Invoice Dashboard	8766	net.systack.invoice-dashboard	✓ Running
Customer Portal	8768	net.systack.customer-dashboard	✓ Running
Orchestrator	N/A	net.systack.orchestrator	✓ Running
Webhook Bridge	8767	net.systack.webhook-bridge	✓ Running

3. Data Flow

Customer Portal Data Flow

```

Dashboard HTML (index.html)
  ↓ fetch /api/auth/login (POST client_id + PIN)
Customer Portal API (api.py :8768)
  ↓
PostgreSQL (systack_memory)
  ├── saos_clients (account info, auth_pin)
  ├── client_auth_tokens (session tokens, expiry)
  ├── agent_state (agent status)
  ├── task_queue (task history)
  ├── chat_conversations (chat threads)
  ├── chat_messages (message history)
  └── message_bus (inter-agent comms)

```

Authentication Flow

```

User (Browser)
  ↓ POST /api/auth/login
  {client_id: int, pin: string}
  ↓
API validates PIN against saos_clients.auth_pin
  ↓
Generates secure random token (secrets.token_urlsafe)
  ↓
Stores token hash (SHA256) in client_auth_tokens
  ↓
Returns Bearer token to client
  ↓
Client stores in localStorage
  ↓
All subsequent requests: Authorization: Bearer <token>
  ↓
API validates token hash, checks expiry, revokes on logout

```

Chat Flow

```

Client Dashboard
  ↓ POST /api/chat/conversations
  ↓ GET /api/chat/conversations/{id}/messages
  ↓ POST /api/chat/conversations/{id}/messages
Customer Portal API
  ↓
PostgreSQL chat_conversations + chat_messages

```

```
↓
Polling: GET /api/chat/poll?since={timestamp}
↓
Agent responses via webhook: POST /api/chat/webhook/agent-response
```

Provisioning Pipeline

```
Stripe Checkout
↓
n8n Webhook (test/production)
↓
saos_provision_bridge.py (:8767)
↓
PostgreSQL task_queue (DEPLOY task)
↓
Orchestrator polls every 10s
↓
ASSEMBLY executes provision_vps.py
↓
Vultr API → VPS created
↓
Client email sent
```

4. Mobile Architecture

Mobile-First Design Principles

FEATURE	IMPLEMENTATION
Responsive breakpoints	768px (tablet), 480px (mobile)
Touch targets	Minimum 44px × 44px
Safe areas	<code>env(safe-area-inset-*)</code> for notches
Navigation	Hamburger menu (≡) on mobile
Sidebar	Slide-out overlay on mobile
Input	<code>inputmode="numeric"</code> for PIN, <code>font-size: 16px</code> to prevent iOS zoom
Form validation	JavaScript-only (button type="button"), no HTML5 validation

Mobile Authentication

```
iOS/Android
↓
Tailscale VPN connection
↓
HTTPS via Tailscale TLS (trusted on device)
↓
Dashboard HTML served from Flask (static_folder)
↓
JavaScript detects /dashboard path prefix for API calls
↓
PIN entry → fetch → Bearer token → localStorage
```

↓
Session persists until logout or 30-day expiry

Mobile API Path Resolution

```
Tailscale serve configuration:
/          → proxy http://127.0.0.1:18789 (Systack site)
/dashboard → proxy http://127.0.0.1:8768 (Customer Portal)

Browser at https://host/dashboard/:
fetch('/api/auth/login') → https://host/api/auth/login ❌ WRONG
fetch('/dashboard/api/auth/login') → https://host/dashboard/api/auth/login ✓ CORRECT

Solution: API_BASE = '/dashboard' when path starts with '/dashboard'
```

5. Security Model

Network Security

- **Tailnet-only** — no public internet exposure
- **HTTPS** — TLS termination via Tailscale
- **mTLS** — Tailscale handles device authentication

Dashboard Authentication

- **PIN-based:** 4-digit PIN per client, stored in PostgreSQL
- **Session tokens:** Secure random 32-byte tokens, SHA256 hashed server-side
- **30-day expiry:** Automatic session cleanup
- **Revocation:** Logout clears token from database
- **Local storage:** Tokens stored in browser localStorage (secure context via HTTPS)

Credential Management

- OAuth secrets: Never committed (see RULE 7)
- API keys: Stored in PostgreSQL, not files
- Tailscale keys: Rotated via admin console

6. Monitoring & Health

Health Checks

- `/api/portal/health` — Customer portal API
- `/api/portal/status` — Full fleet status (requires auth)
- `/api/auth/me` — Session validation
- Tailscale serve status — `tailscale serve status`

Logs

SERVICE	LOG LOCATION
Customer Portal	Systack/content/saos/saos-data/logs/customer-dashboard.log
Fleet Dashboard	Systack/content/saos/saos-data/logs/dashboard.log
Invoice Pipeline	Systack/content/saos/saos-data/logs/invoice-pipeline.log

7. Known Issues & TODO

High Priority

- ☐ Vultr API key integration for auto-provisioning
- ☐ Stripe webhook production test

Medium Priority

- ☐ Billing portal integration
- ☐ Agent performance metrics
- ☐ Cost tracking dashboard
- ☐ iOS Safari certificate trust for .ts.net URLs

Completed

- ☒ Customer portal static file serving
- ☒ Tailscale serve configuration
- ☒ LaunchAgent auto-restart
- ☒ 10-agent fleet documentation
- ☒ PIN-based authentication
- ☒ Session token management
- ☒ Mobile-responsive layout
- ☒ Chat system with conversations
- ☒ Activity audit trail

Document History

VERSION	DATE	CHANGES
1.0	2026-06-17	Initial architecture
2.0	2026-06-23	Added customer portal, 10-agent fleet, Tailscale serve
3.0	2026-06-25	Added PIN authentication, session tokens, mobile architecture section, API path resolution, chat system

Latest version: Internal wiki